UNIVERSITY OF NIAGARA FALLS

Master of Data Analytics

Assignment 3: Advanced Analysis and Reflection on Skills Development

Course: Spring 2025 Advanced Data Visualization (CPSC-600-4)

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Introduction

In the process of completing these three assignments, we have gained valuable technical and analytical skills directly aligned with the demands of today's data analytics industry. This reflection outlines the specific techniques we have acquired such as dataset blending and the use of advanced Tableau functionalities and explores how these skills enhance the readiness for a professional role in data analytics.

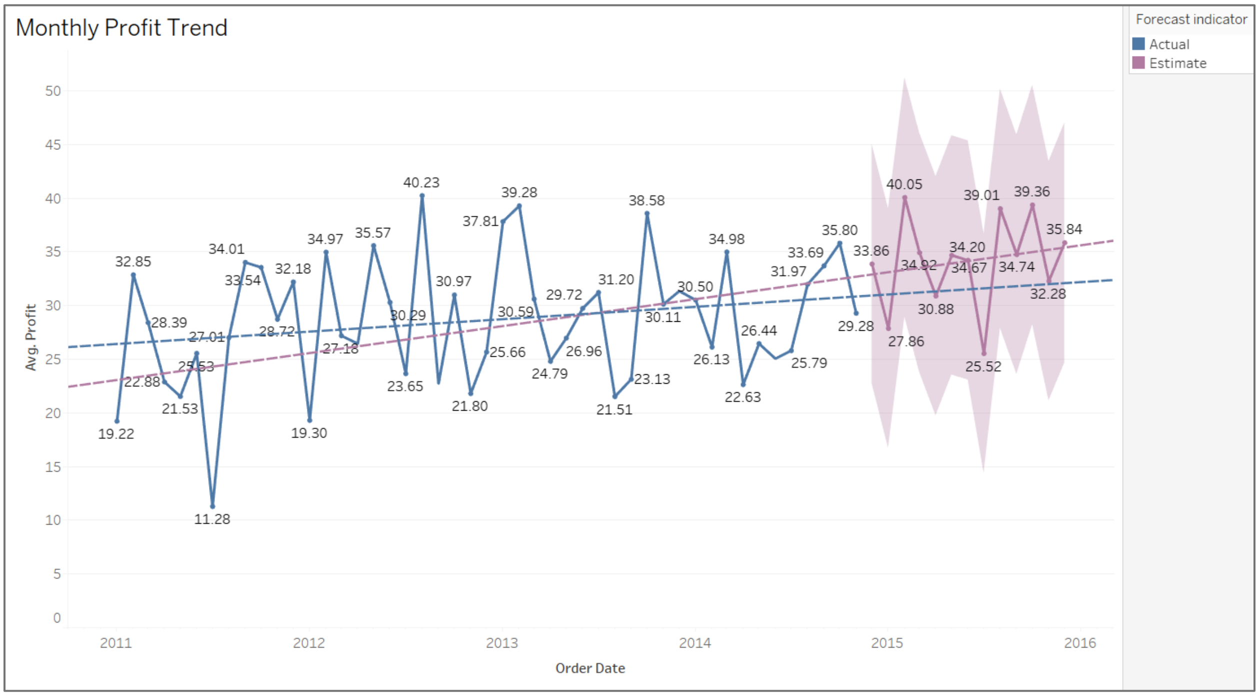
Advanced Tableau Features

To explore deeper insights from the Superstore dataset, advanced Tableau features were applied, including trend lines, forecasting, and calculated fields. These tools enabled a more accurate interpretation of business performance and helped identify emerging patterns. The following visualizations illustrate how these techniques were used to support data-driven decisions.

Predictive analytics tools like trend lines or forecasting

Figure 1

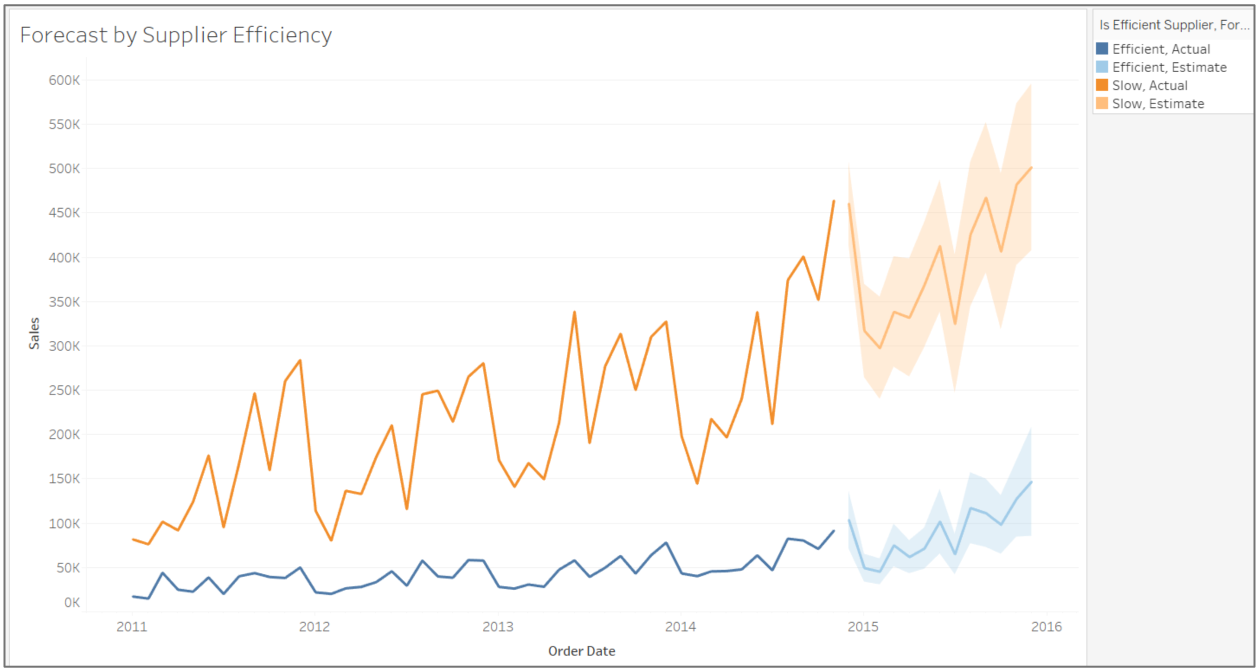
Monthly Profit Trend.



Note. This line chart shows the average monthly profit over time, along with a forecast for the upcoming months. Although the profit fluctuates frequently, the trend line reveals a gradual upward movement. The forecast suggests continued growth with some variability, indicating a generally stable but dynamic business environment.

Figure 2

Forecast by Supplier Efficiency.

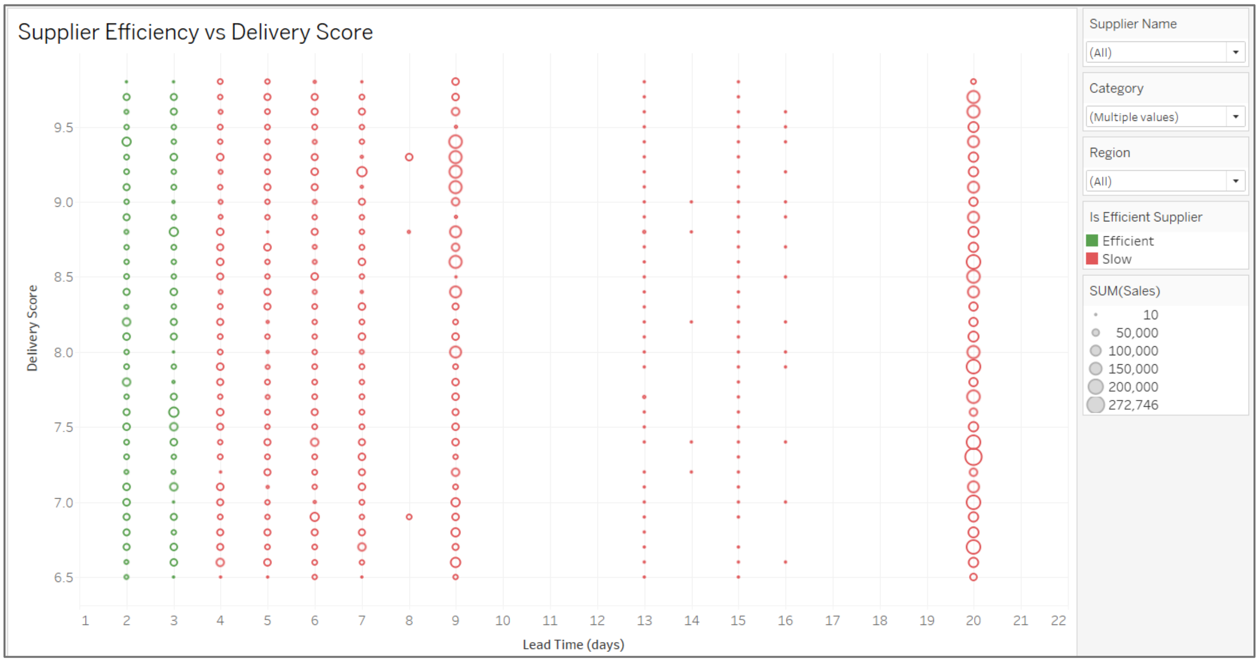


Note. This visualization compares historical and projected sales based on supplier efficiency. Despite being slower, suppliers classified as "Slow" have consistently driven higher sales volumes over time. The forecast suggests that this trend will continue, with both groups showing growth but a much larger projected increase from slower suppliers. This insight highlights a potential trade-off between speed and revenue, suggesting that efficiency alone may not determine sales performance.

New Visualizations

Figure 3

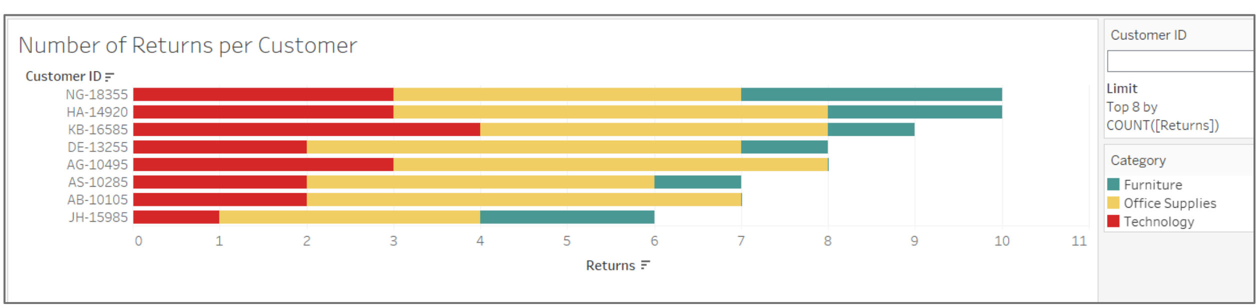
Supplier Efficiency vs Delivery Score.



Note. This scatter plot compares suppliers based on their lead time and delivery score. Efficient suppliers (in green) tend to have shorter delivery times, usually under 5 days, and maintain high delivery scores. However, many slow suppliers (in red) also receive high delivery scores despite their longer lead times. This suggests that customers may value reliability and quality over speed alone. The size of each circle indicates total sales, revealing that top-performing suppliers are not necessarily the fastest ones.

Figure 4

Number of Returns per Customer.

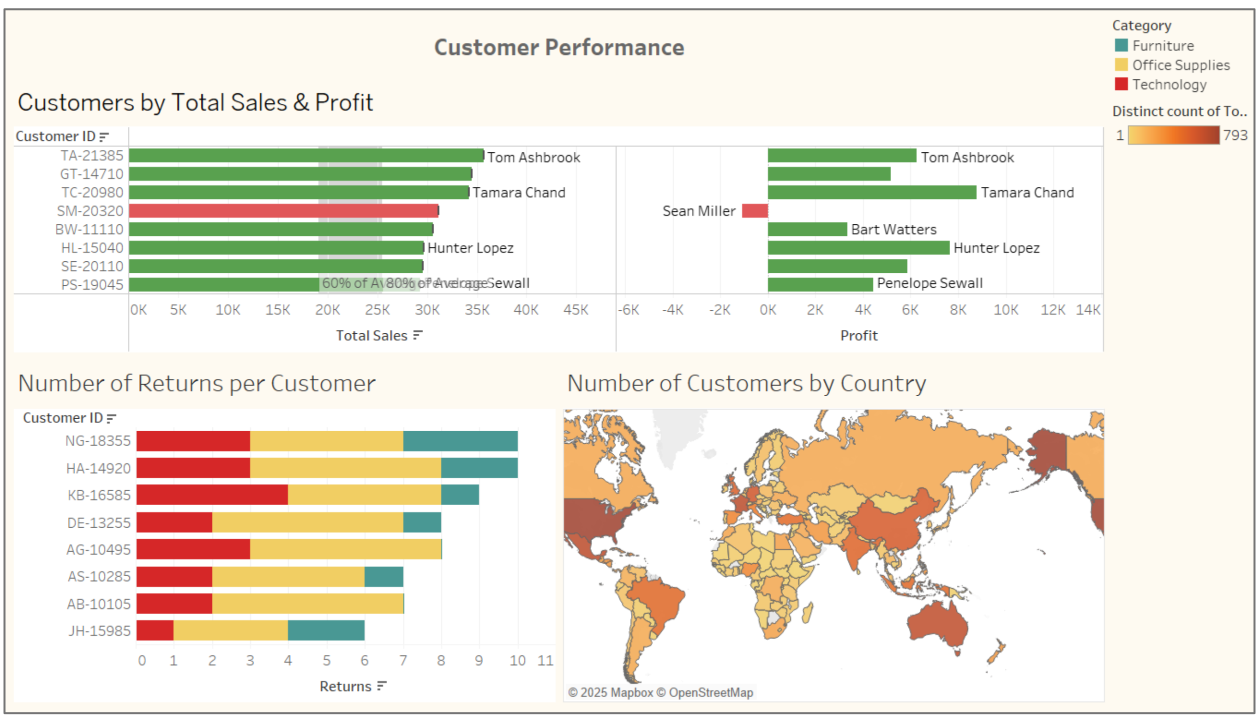


Note. This horizontal bar chart highlights the top eight customers with the highest number of product returns, broken down by product category. Most returns are concentrated in the Office Supplies and Technology categories. While some customers return a variety of products, others consistently return items from a single category. This information can help target follow-ups with high-return customers or investigate quality issues in specific product lines.

Interactive Dashboard

Figure 5

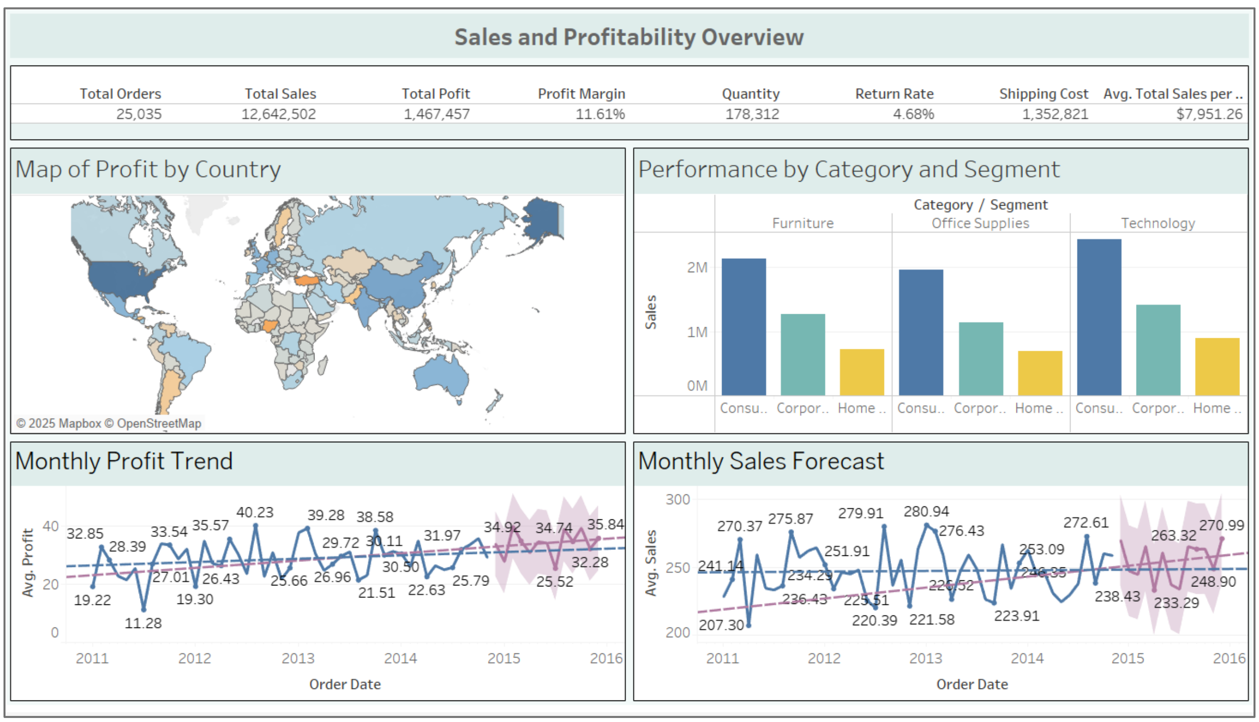
Dashboard Customer Performance.



Note. This dashboard provides a comprehensive view of customer behavior. The top section compares sales and profit by customer, revealing that while most high-sale customers are profitable, a few, like Sean Miller, generate losses despite high revenue. The lower left panel highlights customers with the highest number of returns, many of whom also appear in the sales chart. On the right, the map shows that customer distribution is strongest in North America, Western Europe, and parts of Asia. Together, these visuals help identify valuable customers, regions with high customer concentration, and potential risks from high-return clients.

Figure 6

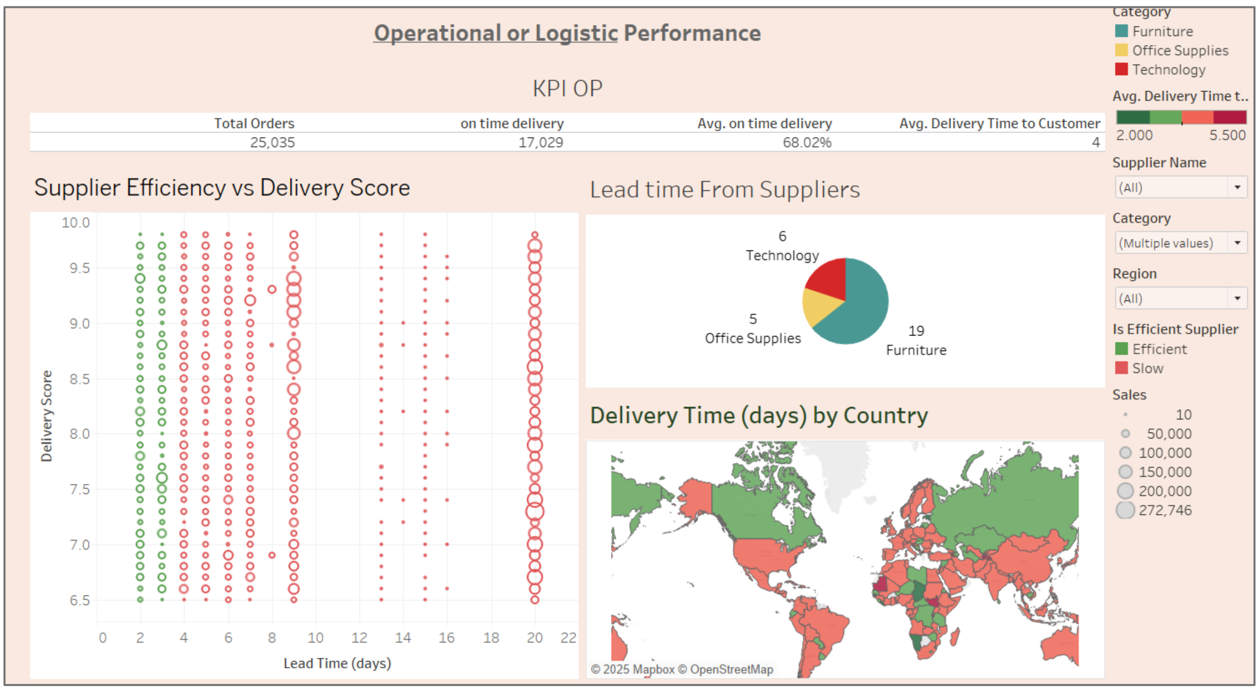
Dashboard Sales and Profitability Overview.



Note. This dashboard provides a high-level summary of sales performance. The top KPIs show over 25,000 orders and $12.6M in sales, with an average profit margin of 11.6% and a return rate under 5%. The profit map indicates strong performance in North America and Western Europe, while the sales bar chart reveals that the Technology category leads across all segments, especially among consumers. At the bottom, time series analyses display positive trends in both profit and sales, with forecasts suggesting continued growth. Together, these insights provide a solid foundation for strategic planning and performance evaluation.

Figure 7

Operational or Logistic Performance.



Note. This dashboard focuses on supplier and delivery performance. With an on-time delivery rate of 68% and an average delivery time of 4 days, there is room for improvement in operational efficiency. The scatter plot shows that while efficient suppliers (green) maintain short lead times and high delivery scores, many slow suppliers (red) also achieve good evaluations despite longer delivery times. The supplier distribution reveals that most are concentrated in the Furniture category. The map at the bottom indicates that delivery performance varies widely by country, highlighting global logistical challenges. These insights are critical for improving supply chain reliability and customer satisfaction.

Reflection

Skills and Techniques Acquired

Throughout this assignment series, we developed key skills in data analysis and visualization using Tableau. We started by building dashboards with basic charts and progressively moved to more advanced features such as calculated fields, parameters, Level of Detail (LOD) expressions, trend lines, and forecasting. One of the most valuable experiences was blending datasets—specifically integrating supplier and returning data with the Superstore dataset. This allowed us to perform a deeper analysis across multiple dimensions, including logistics, customer behavior, and profitability.

We also strengthened our ability to create interactive dashboards. By adding dynamic filters and parameter-driven visualizations, we enhanced the user experience and gave decision-makers more control over the data. We learned to design dashboards not only to look visually appealing, but to answer specific business questions.

Career Readiness and Industry Alignment

The techniques we applied in this assignment reflect real-world expectations for data analysts. Employers often require skills in data preparation, dashboard design, and predictive analytics. Tableau is a widely used tool in the industry, and this project helped us gain practical experience using it in realistic business scenarios. The ability to create clean, focused, and actionable dashboards is a core competency in the job market, and this assignment helped us meet that standard.

Working with multi-source data, customizing calculations, and delivering insights through storytelling are all valuable skills aligned with what the industry expects from entry-level to advanced analysts. We now feel more confident about applying these tools and approaches in professional settings.

Real-World Applications and Business Value

Each dashboard we developed in this assignment answered relevant business questions. For example, we used visualizations to identify customers with high return rates, suppliers with long lead times, and regional sales trends. Forecasting sales and profits added value by supporting strategic planning and performance projections.

These visualizations are not only technically sound but also help drive decisions. For instance, companies could use this analysis to target improvements in supplier performance, reduce costs associated with returns, or realign their marketing based on customer segments and regions. The dashboards demonstrated how visual analytics can improve clarity, reduce complexity, and enable better decisions across departments.

Conclusion

This project series gave us a hands-on foundation in Tableau while also sharpening our analytical thinking and storytelling skills. We learned to transform raw data into business insights, and to design dashboards that are interactive, predictive, and valuable. As a team, we are better prepared for roles in data analytics and confident in our ability to apply what we’ve learned in real-world contexts.